The cross-rank commutator $[R_e(t), R_f(t)]$ in closed form

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Title: The cross-rank commutator $[R_e(t), R_f(t)]$ in closed form
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Annotation, 8 September 2026: the clause "$e = f \Rightarrow$ the two-bead element is 0" in Thm. 2.3(2) is a *one-parameter* statement. It is true as stated (both operators at the same t) and *false* in the two-parameter generality: at $e = f$ there are two legal assignments whose indices are negatives of one another, and they cancel only on $s = t$. See `2026-09-07-c2-Q96-order-over-sublattice.tex` §4.

Abstract

Let $R_e(t)$ be the operator on the ring of symmetric functions that adds a connected e-ribbon with weight t^{ht}. We compute the commutator $[R_e(t), R_f(t)]$ for all $e, f \geq 1$ in closed form. It has exactly two pieces, separated by how many beads of the Maya diagram they transport:

$$[R_e(t), R_f(t)] \;=\; -\frac{1+t}{t}\Big(\Phi_{e,f} \;+\; (t-1)\,\Psi_{e,f}\Big),$$

where $\Phi_{e,f}$ is a one-bead operator — a dressed $(e+f)$-ribbon move — and $\Psi_{e,f}$ is a genuine two-bead operator. Both are written explicitly as undeformed Clifford expressions.

Four consequences. (i) The divisibility of $[R_e, R_f]$ by $(1+t)$, previously proved by evaluating at the anchor $t = -1$, becomes *structural*: every matrix element is a difference of two routes whose ht-statistics differ by exactly 1, so the element is a multiple of $t^{N-1}(1+t)$ before any cancellation. (ii) $\Psi_{e,f} = 0$ if and only if $\min(e, f) = 1$; consequently $[R_1(t), R_f(t)]$ — Zabrocki's add-one-cell operator — is purely one-bead and we give it in closed form. (iii) The one-bead coefficient has a clean combinatorial reading: it compares whether the ribbon *turns* at position e against whether it turns at position f. (iv) The conjecture the session set out to test is **false**: $\partial_t[R_e, R_f]\big|_{t=-1}$ is not a multiplication operator, so it is not a Lie bracket on the span of the power sums; the family $\{R_e(t)\}$ spans no Lie algebra, and the Lie algebra it generates contains operators of bead-number 3.

Two gaps in the first draft are closed in §7: $\Psi_{e,f} \neq 0$ for *all* $e, f \geq 2$, by a uniform witness; and the three-bead sector of $[R_e, [R_f, R_g]]$ factorises as $t^P\big(t^{\kappa_{32}} - t^{-\kappa_{32}}\big)\big(t^{\kappa_{21}+\kappa_{31}} - t^{-(\kappa_{21}+\kappa_{31})}\big)$, which makes the bead-3 statement uniform. What remains open is stated in §8; the honest headline there is that this session had no network, so the $W_{1+\infty}$ / Bloch–Okounkov novelty question is *unchecked* and nothing here claims priority.

Verified by three code-disjoint engines: 1005/1005 (combinatorial vs. Clifford, $|\lambda| \leq 8$, $1 \leq e < f \leq 6$) and 750/750 (combinatorial vs. route-counting, including $e = f$).

1 Conventions and what is assumed

We keep the conventions of [1] verbatim.

Convention 1. Λ is the ring of symmetric functions over $\mathbb{Q}(t)$ with Schur basis $\{s_\lambda\}$ and Hall inner product. For a partition λ the *Maya set* is $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$; its elements are *beads*, its complement *holes*, and $m(x) = [\, x \in M\,]$. We write $|\lambda\rangle$ for s_λ and identify Λ with the charge-0 semi-infinite wedge $\mathcal{F}$ spanned by the $|M\rangle$, with undeformed free fermions $\{\psi_a, \psi_b^*\} = \delta_{ab}$ and $n_a = \psi_a \psi_a^*$. For $e \geq 1$,

$$R_e(t)\, s_\lambda = \sum_\mu t^{\mathrm{ht}(\mu/\lambda)} s_\mu,$$

the sum over $\mu \supset \lambda$ with μ/λ a connected border strip of size e and $\mathrm{ht} = (\#\text{rows})-1$. We abbreviate $N_{(x,y)} = \sum_{x<j<y} n_j$ and $\#(M \cap (x,y))$ for its eigenvalue.

We use one input, proved in [1, Lem. 2.3, Lem. 2.4, Thm. 3.3]:

Proposition 2 (abacus form of R_e; [1])**.** *Call the pair (b,g) a* legal g-move *on M if $b \in M$ and $b+g \notin M$, and write $M \cdot (b,g) := \big(M \setminus \{b\}\big) \cup \{b+g\}$. Then*

$$R_g(t)\, |M\rangle = \sum_{(b,g)\ \text{legal}} t^{\,\#(M\cap(b,b+g))}\, \big|M\cdot(b,g)\big\rangle,$$

a finite sum; the map $b \mapsto M \cdot (b,g)$ is a bijection from legal g-moves onto the μ with μ/λ a connected g-ribbon, and $\mathrm{ht}(\mu/\lambda) = \#(M \cap (b, b+g))$. Equivalently, as operators, $R_g(t) = \sum_{b\in\mathbb{Z}} \psi_{b+g}\psi_b^*\, (-t)^{N_{(b,b+g)}}$.

Convention 3 (bead-number grading)**.** An operator X on $\mathcal{F}$ has *bead-number r* if $\langle M'|X|M\rangle = 0$ whenever $|M \triangle M'| \neq 2r$. Since $\{|M\rangle\}$ is a basis, every operator with locally finite matrix decomposes uniquely as $\sum_{r\geq 0} X^{(r)}$ with $X^{(r)}$ of bead-number r. By Proposition 2 each R_g has bead-number 1, so a product R_eR_f has bead-number parts only in degrees $r \leq 2$.

2 The matrix elements

Everything follows from an exhaustive count of the ways a product of an e-move and an f-move can reach a given target. This section uses no fermions.

Lemma 4 (the two interference indices)**.** *Fix $e, f \geq 1$ and $b, c \in \mathbb{Z}$, and put $x = c - b$. Set*

$$\alpha = [\, c \in (b, b+e)\,], \quad \beta = [\, c+f \in (b,b+e)\,], \quad \alpha' = [\, b \in (c, c+f)\,], \quad \beta' = [\, b+e \in (c,c+f)\,],$$

and $k^{e,f}_{b,c} := \beta' - \alpha'$. Then

$$(\beta - \alpha) + (\beta' - \alpha') \;=\; [\, x = 0\,] - [\, x = e-f\,].$$

In particular $\beta - \alpha = -k^{e,f}_{b,c}$ unless $b = c$ or $b + e = c + f$.

Proof. In the variable x the four conditions read $\alpha = [0 < x < e]$, $\beta = [-f < x < e-f]$, $\alpha' = [-f < x < 0]$, $\beta' = [e - f < x < e]$. The intervals $(0,e)$ and $(-f, 0)$ together with the point 0 partition the integers of $(-f, e)$; so do $(-f, e-f)$ and $(e-f, e)$ together with the point $e-f$. Hence $\alpha + \alpha' + [x = 0] = [\, -f < x < e\,] = \beta + \beta' + [x = e - f]$, which is the claim. $\square$

The number $k^{e,f}_{b,c}$ has a direct meaning: it is the change in $\#(M\cap(c, c+f))$ caused by performing the e-move at b — $+1$ if the moving bead lands inside the interval $(c, c+f)$, -1 if it leaves it, 0 otherwise.

Theorem 5 (matrix elements of the cross-rank commutator)**.** *Let $e, f \geq 1$, let M be a Maya set, and let $M' \neq M$. Then $\langle M'|[R_e, R_f]|M\rangle = 0$ unless $|M \triangle M'| \in \{2, 4\}$, and:*

(1) One-bead sector. *If $M' = M \cdot (b, e+f)$ is a legal $(e+f)$-move, write $N = \#(M \cap (b, b+e+f))$. Then*

$$\boxed{\langle M'|[R_e, R_f]|M\rangle = \big(m(b+e) - m(b+f)\big)\,(1+t)\,t^{N-1}.}$$

(The exponent $N-1$ is never negative on the support: if $N = 0$ then $m(b+e) = m(b+f) = 0$ and the coefficient vanishes.)

(2) Two-bead sector. *Suppose $|M \triangle M'| = 4$. If $e = f$ then $\langle M'|[R_e, R_f]|M\rangle = 0$. If $e \neq f$, then the element vanishes unless $M' = \big(M \setminus \{b, c\}\big) \cup \{b+e, c+f\}$ for a (then unique) pair b, c with $b, c \in M$, $b+e, c+f \notin M$ and $b, c, b+e, c+f$ pairwise distinct. In that case, with $P = \#(M \cap (b, b+e))$, $Q = \#(M \cap (c, c+f))$ and $k = k^{e,f}_{b,c}$,*

$$\boxed{\langle M'|[R_e, R_f]|M\rangle = t^{P+Q}\big(t^{-k} - t^{k}\big).}$$

(Again no negative exponent occurs: $k = 1$ forces $P \geq 1$ and $k = -1$ forces $Q \geq 1$.)

Proof. That only $|M \triangle M'| \in \{0, 2, 4\}$ occurs is Convention 3; the case $M' = M$ is excluded because a route adds $e + f > 0$ cells, so $|\mu'| = |\mu| + e + f$.

By Proposition 2,

$$\langle M'|R_f R_e|M\rangle = \sum_{\text{routes}} t^{w_1+w_2}, \tag{1}$$

summed over pairs of legal moves $M \xrightarrow{(x,e)} M_1 \xrightarrow{(y,f)} M'$, with $w_1 = \#(M \cap (x, x+e))$ and $w_2 = \#(M_1 \cap (y, y+f))$. (Note $R_f R_e$ applies R_e first.) We enumerate the routes.

Step 1: which routes cancel. We have $M' = \big(M \setminus \{x\} \cup \{x+e\}\big) \setminus \{y\} \cup \{y+f\}$. The four sites x, y (removed) and $x+e, y+f$ (added) leave $|M \triangle M'| = 4$ unless one removed site coincides with one added site. Since $x \neq x+e$ and $y \neq y+f$, the only possibilities are $y = x+e$ or $x = y+f$.

Step 2: the one-bead sector. Let $M' = M \cdot (b, e+f)$, so $b \in M$, $b+e+f \notin M$.

Route B ($y = x+e$). Then $M' = M \setminus \{x\} \cup \{x+e+f\}$, forcing $x = b$. Legality: $b \in M$ and $b+e \notin M$ for the first move; for the second, $y = b+e \in M_1$ holds automatically and $y+f = b+e+f \notin M_1$ is equivalent to $b+e+f \notin M$, which is given. So route B exists iff $m(b+e) = 0$. Its weight: $w_1 = \#(M \cap (b, b+e))$, and since the two sites at which M_1 differs from M, namely b and $b+e$, both lie outside the *open* interval $(b+e, b+e+f)$, we get $w_2 = \#(M \cap (b+e, b+e+f))$. Hence $w_1 + w_2 = \#(M \cap (b, b+e+f)) - m(b+e) = N$, using $m(b+e) = 0$.

Route C ($x = y+f$). Then $M' = M \setminus \{y\} \cup \{y+e+f\}$, forcing $y = b$ and $x = b+f$. Legality: $x = b+f \in M$ is required; $x+e = b+e+f \notin M$ is given; $y = b \in M_1$ holds since $b \in M$ and $b \neq b+f$; and $y+f = b+f \notin M_1$ holds because M_1 removed it. So route C exists iff $m(b+f) = 1$. Its weight: $w_1 = \#(M \cap (b+f, b+e+f))$, and since M_1 differs from M only at $b+f$ and $b+e+f$, both outside the open interval $(b, b+f)$, we get $w_2 = \#(M \cap (b, b+f))$. Hence $w_1 + w_2 = \#(M \cap (b, b+e+f)) - m(b+f) = N - 1$, using $m(b+f) = 1$.

Routes B and C are mutually exclusive as *route types* only in the sense that $y = x+e$ and $x = y+f$ cannot hold simultaneously ($e+f = 0$); both may nevertheless occur, for different x. Summing (1),

$$\langle M'|R_f R_e|M\rangle = \big(1 - m(b+e)\big)t^{N} + m(b+f)\,t^{N-1}.$$

Exchanging the roles of e and f (the target $M \cdot (b, e+f)$ is unchanged) gives

$$\langle M'|R_e R_f|M\rangle = \big(1 - m(b+f)\big)t^{N} + m(b+e)\,t^{N-1},$$

and subtracting,

$$\langle M'|[R_e, R_f]|M\rangle = \big(m(b+e) - m(b+f)\big)\big(t^N + t^{N-1}\big),$$

which is (1). For the parenthetical: $b+e$ and $b+f$ both lie in the open interval $(b, b+e+f)$ because $e, f \geq 1$; so $N = 0$ forces $m(b+e) = m(b+f) = 0$.

Step 3: the two-bead sector. Here no cancellation occurs, so by Step 1 the removed sites are $\{x, y\} = M \setminus M'$ and the added sites are $\{x+e, y+f\} = M' \setminus M$, all four distinct. Write $M \setminus M' = \{b, c\}$. If the e-move were made from c, then $c + e \in \{b+e, c+f\}$ would force $c = b$ or $e = f$. So for $e \neq f$ the assignment $x = b$, $y = c$ is forced once we name b as the bead displaced by e; the pair (b, c) is determined by M, M' because b is the unique element of $M \setminus M'$ with $b+e \in M' \setminus M$ (the alternative $b \mapsto c+f$, $c \mapsto b+e$ would give $c - b = f - e$ and $c - b = e - f$ simultaneously, hence $e = f$).

Legality: $b \in M$, $b + e \notin M$; then $c \in M_1$ iff $c \in M$ (as $c \neq b$, $c \neq b+e$), and $c + f \notin M_1$ iff $c + f \notin M$ (as $c+f \neq b$, $c+f \neq b+e$). Weights: $w_1 = P$, and

$$w_2 = \#(M_1 \cap (c, c+f)) = Q - [\, b \in (c, c+f)\,] + [\, b+e \in (c, c+f)\,] = Q + k.$$

So $\langle M'|R_f R_e|M\rangle = t^{P+Q+k}$. Running the same argument with e and f exchanged — now the f-move at c comes first — gives weights Q and $P + (\beta - \alpha)$, and $\beta - \alpha = -k$ by Lemma 4, since $b \neq c$ and $b + e \neq c + f$. So $\langle M'|R_e R_f|M\rangle = t^{P+Q-k}$ and the difference is as claimed.

If $e = f$ the two assignments $(x, y) = (b, c)$ and $(x, y) = (c, b)$ are both available in each order, and the two orders contribute the same multiset of weights, so the commutator vanishes; this also follows from $[R_e, R_e] = 0$.

Finally the sign conditions on exponents. If $k = 1$ then $b+e \in (c, c+f)$ and $b \notin (c, c+f)$; from $b < b+e < c+f$ and $b \notin (c, c+f)$ we get $b \leq c$, hence $b < c < b+e$, so $c \in M \cap (b, b+e)$ and $P \geq 1$. If $k = -1$ then $b \in (c, c+f)$, so $b \in M \cap (c, c+f)$ and $Q \geq 1$. $\square$

3 The closed operator form

Theorem 6 (cross-rank commutator)**.** *For all $e, f \geq 1$, as operators on $\mathcal{F} \cong \Lambda$,*

$$\boxed{[R_e(t), R_f(t)] \;=\; -\frac{1+t}{t}\Big(\Phi_{e,f} + (t-1)\,\Psi_{e,f}\Big)}$$

where

$$\Phi_{e,f} \;=\; \sum_{b \in \mathbb{Z}} \psi_{b+e+f}\, \psi_b^*\, (-t)^{N_{(b,\, b+e+f)}}\, \big(n_{b+f} - n_{b+e}\big)$$

is a bead-number-1 operator (a dressed $(e+f)$-ribbon move), and

$$\Psi_{e,f} \;=\; \sum_{b,c \in \mathbb{Z}} k_{b,c}^{e,f}\, \psi_{b+e}\, \psi_{c+f}\, \psi_b^*\, \psi_c^*\, (-t)^{N_{(b,\, b+e)} + N_{(c,\, c+f)}}$$

is a bead-number-2 operator, with $k_{b,c}^{e,f}$ as in Lemma 4. Both sums are locally finite.

Proof. Write $A_b = \psi_{b+e}\psi_b^*$, $C_c = \psi_{c+f}\psi_c^*$, $D_b = (-t)^{N_{(b,b+e)}}$, $G_c = (-t)^{N_{(c,c+f)}}$, so that $R_e = \sum_b A_b D_b$ and $R_f = \sum_c C_c G_c$ by Proposition 2.

Step 1: moving the dressings across. C_c transports a bead from c to $c+f$, so on its support the eigenvalue of $N_{(b,b+e)}$ changes by $\beta - \alpha$ in the notation of Lemma 4; off its support both sides below vanish. Hence, as operators,

$$D_b C_c = (-t)^{\beta-\alpha}\, C_c D_b, \qquad \text{and symmetrically} \qquad G_c A_b = (-t)^{\beta'-\alpha'}\, A_b G_c.$$

Since D_b and G_c are both diagonal they commute, so

$$A_b D_b\, C_c G_c = u_{bc}\, A_b C_c\, D_b G_c, \qquad C_c G_c\, A_b D_b = v_{bc}\, C_c A_b\, D_b G_c,$$

with $u_{bc} = (-t)^{\beta-\alpha}$ and $v_{bc} = (-t)^{\beta'-\alpha'} = (-t)^{k^{e,f}_{b,c}}$.

Step 2: the Clifford products. Using $\psi_j^*\psi_k = \delta_{jk} - \psi_k\psi_j^*$,

$$A_b C_c = \delta_{b,c+f}\, \psi_{b+e}\psi_c^* - \Theta_{bc}, \qquad C_c A_b = \delta_{c,b+e}\, \psi_{c+f}\psi_b^* - \Theta_{bc},$$

where $\Theta_{bc} = \psi_{b+e}\psi_{c+f}\psi_b^*\psi_c^*$ is the *same* operator in both lines, because $\psi_{c+f}\psi_{b+e}\psi_c^*\psi_b^* = (-\psi_{b+e}\psi_{c+f})(-\psi_b^*\psi_c^*) = \Theta_{bc}$. Therefore

$$[R_e, R_f] = \underbrace{\sum_{b,c}\Big(u_{bc}\delta_{b,c+f}\, \psi_{b+e}\psi_c^* - v_{bc}\delta_{c,b+e}\, \psi_{c+f}\psi_b^*\Big) D_b G_c}_{\text{resonant}} \;-\; \underbrace{\sum_{b,c}(u_{bc} - v_{bc})\, \Theta_{bc}\, D_b G_c}_{\text{quartic}}. \tag{2}$$

Step 3: the resonant part is $-\frac{1+t}{t}\Phi_{e,f}$. In the first sum $b = c + f$, i.e. $x = c - b = -f$; then $\alpha = [0 < -f < e] = 0$ and $\beta = [-f < -f < e - f] = 0$, so $u_{bc} = 1$. In the second sum $c = b + e$, i.e. $x = e$; then $\alpha' = [-f < e < 0] = 0$ and $\beta' = [e - f < e < e] = 0$, so $v_{bc} = 1$. Reindexing both by the moving bead b, and using

$$N_{(b,b+e)} + N_{(b+e,b+e+f)} = N_{(b,b+e+f)} - n_{b+e}, \qquad N_{(b+f,b+e+f)} + N_{(b,b+f)} = N_{(b,b+e+f)} - n_{b+f},$$

the resonant part equals

$$\sum_b \psi_{b+e+f}\psi_b^*\, (-t)^{N_{(b,b+e+f)}}\Big[(-t)^{-n_{b+f}} - (-t)^{-n_{b+e}}\Big].$$

(The diagonal operators n_{b+e}, n_{b+f} commute with $\psi_{b+e+f}\psi_b^*$ because $b+e, b+f \notin \{b, b+e+f\}$, as $e, f \geq 1$.) Since $n^2 = n$ we have $(-t)^{-n} = 1 + \big((-t)^{-1} - 1\big)n$, so the bracket equals $-\frac{1+t}{t}\,(n_{b+f} - n_{b+e})$, giving $-\frac{1+t}{t}\Phi_{e,f}$.

Step 4: the quartic part is $-\frac{(t-1)(1+t)}{t}\Psi_{e,f}$. Note first that $\Theta_{bc} = 0$ when $b = c$ (then $\psi_b^*\psi_c^* = 0$) and when $b + e = c + f$ (then $\psi_{b+e}\psi_{c+f} = 0$). Away from these two values of x, Lemma 4 gives $\beta - \alpha = -k^{e,f}_{b,c}$, hence

$$u_{bc} = (-t)^{-k}, \qquad v_{bc} = (-t)^{k}, \qquad u_{bc} - v_{bc} = (-t)^{-k} - (-t)^{k} = k\,\frac{t^2-1}{t}$$

for $k = k^{e,f}_{b,c} \in \{-1, 0, 1\}$ (check the three values directly; note $k \in \{-1, 0, 1\}$ because $\alpha', \beta' \in \{0, 1\}$). Thus the quartic part of (2) equals $-\frac{t^2-1}{t}\Psi_{e,f} = -\frac{(t-1)(1+t)}{t}\Psi_{e,f}$.

Step 5: bead numbers. $\Phi_{e,f}$ visibly has bead-number 1. For $\Psi_{e,f}$: whenever $k^{e,f}_{b,c} \neq 0$ and $b \neq c$, the four sites $b, c, b+e, c+f$ are pairwise distinct. Indeed $b \neq b + e$ and $c \neq c + f$; if $c = b + e$ then $x = e$ and we computed $\alpha' = \beta' = 0$, so $k = 0$; if $b = c + f$ then $x = -f$ and again $\alpha' = \beta' = 0$, so $k = 0$; and if $b + e = c + f$ then $x = e - f$, where $\alpha' = [-f < e - f < 0] = 0$ and $\beta' = [e - f < e - f < e] = 0$, so $k = 0$. The remaining degenerate case $b = c$ has $\Theta_{bc} = 0$. Hence every surviving summand transports two beads.

Local finiteness of both sums is inherited from Proposition 2. Finally, the matrix elements of the right-hand side agree with Theorem 5: this is the consistency check recorded in §6, and it is what pins the fermionic signs. $\square$

Corollary 7 (structural divisibility)**.** *$(1+t)$ divides every matrix element of $[R_e(t), R_f(t)]$, in $\mathbb{Z}[t]$. More precisely, in the one-bead sector every element is $\pm(1+t)t^{N-1}$ or 0, and in the two-bead sector $\pm(1-t^2)t^{P+Q-1}$ or 0.*

Proof. Immediate from Theorem 5, since $t^{-k} - t^k = \mp(t^2-1)t^{-1}$ for $k = \pm 1$. $\square$

Remark 8 (why the $(1+t)$ is there)**.** In [1, Cor. 5.6] this divisibility was proved by evaluating at the anchor: $R_e(-1) = M_{p_e}$, multiplications commute, so every coefficient vanishes at $t = -1$. That argument is correct but tells one nothing about the quotient. Theorem 5 replaces it by a mechanism. In the one-bead sector there are exactly two routes from λ to μ, the "in-order push" $b \to b+e \to b+e+f$ and the "leapfrog" $b+f \to b+e+f$ followed by $b \to b+f$; their ht-statistics are N and $N-1$, differing by exactly 1; and exchanging e and f exchanges *which* route is available while preserving the pair of exponents $\{N-1, N\}$. The commutator is therefore forced to be a multiple of $t^{N-1}+t^N$. The same holds in the two-bead sector with the pair $\{P+Q-1, P+Q+1\}$, whence the factor $t^{-k}-t^k$. The divisibility is not a coincidence at a special point; it is the statement that reordering the two ribbons costs exactly one unit of height.

4 The case $e = 1$: Zabrocki's add-one-cell operator

Corollary 9. *$\Psi_{e,f} = 0$ if and only if $\min(e, f) = 1$. Consequently, for every $f \geq 1$, $[R_1(t), R_f(t)]$ has bead-number 1 and*

$$[R_1(t), R_f(t)] \;=\; -\frac{1+t}{t} \sum_{b\in\mathbb{Z}} \psi_{b+f+1}\, \psi_b^*\, (-t)^{N(b,\, b+f+1)} \big(n_{b+f} - n_{b+1}\big),$$

equivalently, on partitions,

$$[R_1(t), R_f(t)]\, s_\lambda = (1+t) \sum_{\mu} \big(m(b+1) - m(b+f)\big)\, t^{\,\mathrm{ht}(\mu/\lambda)-1}\, s_\mu,$$

the sum over μ with μ/λ a connected $(f{+}1)$-ribbon, b its bead move.

Proof. Take $e = 1$. A summand of $\Psi_{1,f}$ survives only if $k^{1,f}_{b,c} \neq 0$ and $\Theta_{bc} \neq 0$. Now $k = \beta' - \alpha'$ with $\alpha' = [-f < x < 0]$ and $\beta' = [1-f < x < 1]$, where $x = c - b$. So $k = +1$ forces $x = 0$ (the only integer in $(1-f, 1)$ not in $(-f, 0)$), where $\Theta_{bc} = 0$; and $k = -1$ forces $x = 1-f$ (the only integer in $(-f, 0)$ not in $(1-f, 1)$), i.e. $b+e = b+1 = c+f$, where again $\Theta_{bc} = 0$. Hence $\Psi_{1,f} = 0$, and by antisymmetry $\Psi_{f,1} = 0$.

Conversely, if $e, f \geq 2$, take c arbitrary and $b = c+f-1$. Then $x = 1-f \leq -1 < 0$ and $x > -f$, so $\alpha' = 1$; and $b+e = c+f+e-1 > c+f$ since $e \geq 2$, so $\beta' = 0$ and $k = -1$. The four sites are distinct: $b \neq c$ (as $f \geq 2$), $b+e \neq c+f$ (as $e \geq 2$), $c \neq b+e$, $b \neq c+f$. So $\Psi_{e,f}$ has a nonzero summand; that the operator itself is nonzero is Corollary 12 below, proved from Theorem 5.

The displayed formulas are Theorem 6 with $\Psi = 0$, and Theorem 5(1) with $e = 1$ combined with Proposition 2. $\square$

The one-bead coefficient has a purely combinatorial reading. Recall (Proposition 2) that a connected g-ribbon μ/λ corresponds to a bead move $b \to b+g$; read its g cells from the southwest tail to the northeast head and let $\tau_j(\mu/\lambda) \in \{0, 1\}$ record whether the j-th and $(j{+}1)$-st cells lie in *different* rows, for $1 \leq j \leq g-1$.

Lemma 10. *$\tau_j(\mu/\lambda) = m(b+j)$ for $1 \leq j \leq g-1$.*

Proof. This is the cell-by-cell refinement of [1, Lem. 2.4]. In that proof the rows of μ that grow are exactly those attached to the beads of M in the open interval $(b, b+e)$, each by one cell, together with the moved bead's own row; reading the ribbon from its tail, the j-th internal edge is a row change precisely when the site $b+j$ carries such a bead. (Verified independently on 1178 ribbons, $|\lambda| \le 7$, $g \le 6$: see §6.) $\square$

Corollary 11 (the commutator compares two turning positions)**.** *Let μ/λ be a connected $(e+f)$-ribbon. Then*

$$\langle s_\mu,\ [R_e(t), R_f(t)]\, s_\lambda\rangle \ =\ (1+t)\, t^{\,\mathrm{ht}(\mu/\lambda)-1}\left(\tau_e(\mu/\lambda) - \tau_f(\mu/\lambda)\right).$$

That is: the one-bead part of $[R_e, R_f]$ sees only whether the ribbon turns *after its e-th cell against whether it turns after its f-th cell. It vanishes on every ribbon that behaves the same way at both positions.*

Proof. Theorem 5(1) and Lemma 10. $\square$

5 What the family is not

Corollary 12 (the family spans no Lie algebra)**.** *For $e, f \ge 2$ the two-bead part of $[R_e(t), R_f(t)]$ is nonzero; hence $\Psi_{e,f} \ne 0$, and $[R_e, R_f]$ is not a $\mathbb{Q}(t)$-linear combination of bead-number-1 operators. For $e = 1 < f$ the commutator has bead-number 1 but is still not in $\mathrm{span}_{\mathbb{Q}(t)}\{R_g(t)\}$. In all cases with $e \ne f$,*

$$[R_e(t), R_f(t)] \notin \mathrm{span}_{\mathbb{Q}(t)}\{R_g(t) : g \ge 1\}.$$

Proof. For the first claim it suffices to exhibit one nonzero two-bead matrix element. Take $\lambda = \varnothing$, so $M = \mathbb{Z}_{<0}$, and $e = 2$, $f = 3$, $b = -1$, $c = -3$. Then $b, c \in M$; $b + e = 1 \notin M$; $c + f = 0 \notin M$; and the four sites $-1, -3, 1, 0$ are pairwise distinct, so this is a legitimate two-bead target. Here $x = c - b = -2$, $\alpha' = [-3 < -2 < 0] = 1$, $\beta' = [-1 < -2 < 2] = 0$, so $k = -1$. Also $P = \#(M \cap (-1, 1)) = \#\{0\} \cap M = 0$ and $Q = \#(M \cap (-3, 0)) = \#\{-2, -1\} = 2$. Theorem 5(2) gives the coefficient $t^2(t^1 - t^{-1}) = t^3 - t$, on the target $M' = M \setminus \{-1, -3\} \cup \{1, 0\}$, i.e. $\mu = (2, 2, 1)$. Indeed $[R_2, R_3]s_\varnothing$ contains $(t^3 - t)\, s_{(2,2,1)}$, and $(2, 2, 1)/\varnothing$ requires two beads to move. A bead-number-1 operator has no such matrix element.

For $e = 1 < f$: $[R_1, R_f]$ raises the degree $|\lambda|$ by exactly $f + 1$, whereas R_g raises it by g; so in any expression $[R_1, R_f] = \sum_g c_g R_g$ only $g = f + 1$ can have $c_g \ne 0$, and $[R_1, R_f]$ would be a scalar multiple of R_{f+1}. But by Corollary 9 its matrix element at a ribbon vanishes whenever $\tau_1 = \tau_f$, while the corresponding matrix element of R_{f+1} is $t^{\mathrm{ht}} \ne 0$. Explicitly for $f = 2$: $[R_1, R_2]s_\varnothing = (1+t)s_{(2,1)}$, whereas $R_3 s_\varnothing = s_{(3)} + t\, s_{(2,1)} + t^2 s_{(1,1,1)}$; the supports differ. The last display combines the two cases (for $e, f \ge 2$ the span consists of bead-number-1 operators, which the commutator is not). $\square$

Corollary 13 (the classical limit is not a bracket on the power sums)**.** *Write $[R_e, R_f] = (1 + t)\, C_{e,f}(t)$, legitimate by Corollary 7, and $\{p_e, p_f\} := C_{e,f}(-1) = \partial_t [R_e, R_f]\big|_{t=-1}$. Then $\{p_e, p_f\}$ is* ***not*** *a multiplication operator on Λ; in particular it does not lie in $\mathrm{span}\{M_{p_g}\} = \mathrm{span}\{R_g(-1)\}$, and the conjecture that $\{\cdot,\cdot\}$ is a Lie bracket on $\mathrm{span}\{p_e\}$ is false.*

Proof. Take $e = 1$, $f = 2$. From $[R_1, R_2]s_\varnothing = (1 + t)s_{(2,1)}$ and $[R_1, R_2]s_{(1)} = -(1 + t)s_{(2,2)}$ (both computed from the definition) we get $C_{1,2}s_\varnothing = s_{(2,1)}$ and $C_{1,2}s_{(1)} = -s_{(2,2)}$, independently of t.

If $C_{1,2}(-1)$ were multiplication by some $g \in \Lambda$, then applying it to $s_\varnothing = 1$ would give $g = s_{(2,1)}$, whence by Pieri

$$C_{1,2}(-1)\, s_{(1)} = s_{(2,1)}s_{(1)} = s_{(3,1)} + s_{(2,2)} + s_{(2,1,1)} \;\neq\; -s_{(2,2)}.$$

Contradiction. The same witness rules out membership in $\mathrm{span}\{M_{p_g}\}$, every element of which is a multiplication operator. $\square$

Corollary 14 (the bead-number is unbounded on the generated Lie algebra)**.** *Let $\mathfrak{g}_t \subset \mathrm{End}(\Lambda)$ be the Lie algebra generated by $\{R_e(t) : e \geq 1\}$ over $\mathbb{Q}(t)$. Then $\mathfrak{g}_t$ contains elements of bead-number 3; in particular it is not contained in the span of operators of bead-number ≤ 2.*

Proof. Computation: $\big[R_2, [R_3, R_4]\big] s_\varnothing$ contains the term $(-2t^5 + 4t^3 - 2t)\, s_{(3,3,3)}$, and $M((3,3,3))$ differs from $M(\varnothing)$ in six sites. See §6. Corollary 19 upgrades this to a uniform statement with a proof, for infinitely many triples (e, f, g). $\square$

Remark 15 (what survives of the Poisson picture)**.** Corollary 13 refutes the conjecture as posed but not the underlying shape. Setting $\hbar = 1+t$, the algebra generated by the R_e is a deformation of a commutative algebra — at $\hbar = 0$ every R_e becomes M_{p_e} — and $\{\cdot,\cdot\}$ is the associated first-order bracket. What Corollary 13 shows is that the classical phase space is *not* $\mathrm{span}\{p_e\}$: the bracket leaves it immediately, landing in the dressed $\mathfrak{gl}_\infty$ operators $\Phi_{e,f}(-1) = \sum_b \psi_{b+e+f}\psi_b^*(n_{b+f} - n_{b+e})$ and, for $e, f \geq 2$, in genuine two-body operators. Identifying the smallest Poisson algebra containing the p_e and closed under $\{\cdot,\cdot\}$ is the natural next question; by Corollaries 14 and 19 it is not finitely graded by bead-number.

6 Computational verification

Three engines, sharing no code beyond the partition enumerator, and meeting only at the partition↔Maya interface.

- **Engine A** (`engineA.py`) computes $R_e(t)$ from Convention 1 directly: enumerate $\mu \supset \lambda$ with $|\mu/\lambda| = e$, test that μ/λ is edge-connected and contains no 2×2 block, weight by $t^{\#\mathrm{rows}-1}$. No abacus, no fermions.
- **Engine B** (`engineB.py`) evaluates the closed form of Theorem 6 by applying ψ and ψ^* one at a time to a Maya set, with the sign $(-1)^{\#\{x \in M : x > a\}}$ computed explicitly at each step. No border strips.
- **Engine C** (`engineC.py`) evaluates the route-count formulas of Theorem 5 directly. No fermion signs at all.

Results:

comparison	range	result
A vs. B	$\lvert\lambda\rvert \leq 8$, $1 \leq e < f \leq 6$	1005/1005 agree
A vs. C	$\lvert\lambda\rvert \leq 6$, $1 \leq e, f \leq 5$ (incl. $e = f$)	750/750 agree
$\tau_j = m(b+j)$ (Lem. 10)	$\lvert\lambda\rvert \leq 7$, $g \leq 6$	1178/1178 agree

Two further checks, each of which could have failed:

- $\Psi_{1,f} = 0$, *checked from both sides.* Engine B reports empty support for $\Psi_{1,f}$, $2 \le f \le 6$, $|\lambda| \le 6$ — this is the algebraic prediction. Independently, Engine A, which knows nothing of Ψ, reports *zero* nonzero two-bead matrix elements of $[R_1, R_f]$ for $f = 2, 3, 4$, and 54, 64, 150, 70 of them for $(e, f) = (2, 3), (2, 4), (3, 4), (2, 5)$. The two accounts of "where the two-body part is" agree exactly, and they agree on a boundary ($\min(e, f) = 1$) that neither engine was told about.

- $e = f$. The A-vs-C sweep includes $e = f$, where the commutator must vanish. It does, in all 150 such cases. This tests the antisymmetry of $k^{e,f}_{b,c}$ against the symmetry of $\Theta_{bc} D_b G_c$, i.e. Step 5 of Theorem 6, in a regime where every individual summand is nonzero.

A negative control that moves the prediction: the exponent $N-1$ in Theorem 5(1) is not a fitted constant — replacing $t^{N-1}(1+t)$ by $t^N(1+t)$ changes $[R_1, R_2]s_\varnothing$ from $(1+t)s_{(2,1)}$ to $(t+t^2)s_{(2,1)}$, which Engine A refutes at $|\lambda| = 0$. Likewise the $t = -1$ specialisation is *not* used anywhere in the derivation of Theorem 5; symbolic t is carried throughout, so the anchor remains an independent check (all 1005 commutators vanish at $t = -1$, consistent with [1, Cor. 5.6]).

7 Closing the two gaps: uniform witnesses

Both gaps I first recorded turned out to be shallower than they looked, for the same reason: Theorem 5 determines a matrix element by a *single* configuration, so no cancellation is available to hide behind.

Proposition 16 ($\Psi_{e,f} \neq 0$ for all $e, f \ge 2$)**.** *Let $e, f \ge 2$ with $e \neq f$, take $\lambda = \varnothing$ (so $M = \mathbb{Z}_{<0}$), and let μ be the partition with $M(\mu) = \big(\mathbb{Z}_{<0} \setminus \{-1, -f\}\big) \cup \{e-1, 0\}$. Then*

$$\big\langle s_\mu,\, [R_e(t), R_f(t)]\, s_\lambda \big\rangle = t^{\,f} - t^{\,f-2} \;\neq\; 0.$$

In particular $\Psi_{e,f} \neq 0$, and the two-bead part of $[R_e, R_f]$ is nonzero, for every $e, f \ge 2$.

Proof. Put $b = -1$, $c = -f$. Then $b, c \in M$ (both are negative, using $f \ge 1$); $b + e = e - 1 \ge 1$ and $c + f = 0$ are both $\notin M$; and the four sites $-1, -f, e-1, 0$ are pairwise distinct because $f \ge 2$ (so $-f \neq -1$) and $e \ge 2$ (so $e - 1 \ge 1$, hence $e - 1 \notin \{-1, -f, 0\}$). So this is a legal two-bead configuration with target $M(\mu)$.

Its interference index: $\alpha' = [\,c < b < c + f\,] = [-f < -1 < 0] = 1$ since $f \ge 2$, and $\beta' = [\,c < b + e < c + f\,] = [-f < e - 1 < 0] = 0$ since $e - 1 \ge 1 > 0$. Hence $k^{e,f}_{b,c} = -1$. Also $P = \#\big(M \cap (-1, e-1)\big) = 0$ (that interval contains only non-negative integers) and $Q = \#\big(M \cap (-f, 0)\big) = f - 1$.

By Theorem 5(2) — and this is the point — the pair (b, c) is *determined* by M and $M(\mu)$ when $e \neq f$, so the matrix element is this single term and nothing can cancel it: $t^{P+Q}(t^{-k} - t^k) = t^{f-1}(t - t^{-1}) = t^f - t^{f-2}$. Since $\Phi_{e,f}$ has bead-number 1 it contributes nothing here, so $\Psi_{e,f} \neq 0$. $\square$

Remark 17. For $(e, f) = (2, 3)$ this is the witness of Corollary 12: $\mu = (2, 2, 1)$ and the coefficient is $t^3 - t$. Verified against Engine A for all $2 \le e, f \le 7$, $e \neq f$ (30/30).

We now do the same one level up. Fix three legal moves $m_i = (b_i, g_i)$, $i = 1, 2, 3$, on M, with $(g_1, g_2, g_3) = (e, f, g)$ and with the six sites $b_i, b_i + g_i$ pairwise distinct. Write $P_i = \#\big(M \cap (b_i, b_i + g_i)\big)$ and

$$\kappa_{ij} = \big[\, b_i + g_i \in (b_j, b_j + g_j)\,\big] - \big[\, b_i \in (b_j, b_j + g_j)\,\big],$$

the change move i makes to the statistic of move j. By Lemma 4, $\kappa_{ij} + \kappa_{ji} = 0$ (the two degenerate cases $b_i = b_j$ and $b_i + g_i = b_j + g_j$ are excluded by distinctness of the six sites).

Proposition 18 (the three-bead sector factorises)**.** *Let M' satisfy $|M\triangle M'| = 6$. Then*

$$\langle M'|\,[R_e,[R_f,R_g]]\,|M\rangle = \sum t^{\,P_1+P_2+P_3}\Big(t^{\kappa_{32}} - t^{-\kappa_{32}}\Big)\Big(t^{\,\kappa_{21}+\kappa_{31}} - t^{-(\kappa_{21}+\kappa_{31})}\Big),$$

the sum over the assignments of the three beads of $M\setminus M'$ to the three displacements e,f,g that carry them onto $M'\setminus M$. Each summand vanishes if and only if $\kappa_{32}=0$ or $\kappa_{21}+\kappa_{31}=0$.

Proof. A product of three one-bead moves reaches a target at Hamming distance 6 only if no two of them contract, i.e. the three moves are as above; the assignment of displacements to beads is what the sum ranges over. All six sites being distinct, every ordering of the three moves is legal: move j finds b_j still occupied and b_j+g_j still empty whatever else has happened. Moreover, when move j is performed after the moves in a set S, its weight exponent is $P_j+\sum_{i\in S}\kappa_{ij}$, because each earlier move i removes a bead at b_i and adds one at b_i+g_i, changing $\#\big(\cdot\cap(b_j,b_j+g_j)\big)$ by exactly κ_{ij}, independently of the other moves. Hence performing the moves in the order σ contributes $t^{\,P_1+P_2+P_3+\sum_{i\prec_\sigma j}\kappa_{ij}}$.

Expanding, and recalling that $R_aR_bR_c$ applies R_c first,

$$[R_e,[R_f,R_g]] = R_eR_fR_g - R_eR_gR_f - R_fR_gR_e + R_gR_fR_e$$

corresponds to the four orders $(3,2,1),(2,3,1),(1,3,2),(1,2,3)$ with signs $+,-,-,+$. Their exponent corrections are, respectively,

$$\kappa_{32}+\kappa_{31}+\kappa_{21},\quad \kappa_{23}+\kappa_{21}+\kappa_{31},\quad \kappa_{13}+\kappa_{12}+\kappa_{32},\quad \kappa_{12}+\kappa_{13}+\kappa_{23}.$$

The first two share $\kappa_{31}+\kappa_{21}$ and the last two share $\kappa_{12}+\kappa_{13}$, so with $P=P_1+P_2+P_3$ the bracket equals

$$t^P\Big(t^{\kappa_{32}} - t^{\kappa_{23}}\Big)\Big(t^{\,\kappa_{21}+\kappa_{31}} - t^{\,\kappa_{12}+\kappa_{13}}\Big),$$

and substituting $\kappa_{23}=-\kappa_{32}$, $\kappa_{12}=-\kappa_{21}$, $\kappa_{13}=-\kappa_{31}$ gives the statement. A factor t^a-t^{-a} vanishes iff $a=0$. $\square$

Corollary 19 (bead-number 3 occurs, uniformly)**.** *Let $e\ge 2$, $f\ge 3$, $g\ge e+2$ with e,f,g distinct and $(e,g)\ne(2,f+1)$. Take $\lambda=\varnothing$ and the moves $b_1=-e$ (by e), $b_2=-1$ (by f), $b_3=1-g$ (by g). Then the assignment is unique, $\kappa_{32}=1$, $\kappa_{21}=-1$, $\kappa_{31}=0$, and*

$$\big\langle s_\mu,\,[R_e,[R_f,R_g]]\,s_\varnothing\big\rangle \;=\; -\,t^{\,e+g-5}\,(t^2-1)^2 \;\ne\; 0,$$

where $M(\mu)=\big(\mathbb{Z}_{<0}\setminus\{-e,-1,1-g\}\big)\cup\{0,f-1,1\}$. Hence the Lie algebra $\mathfrak{g}_t$ generated by $\{R_e(t)\}$ contains elements of bead-number 3, for infinitely many triples (e,f,g).

Proof. Legality and distinctness: the removed sites $-e,-1,1-g$ are all negative (using $g\ge e+2\ge 4$) hence in $M=\mathbb{Z}_{<0}$, and are distinct because $e\ge 2$, and $1-g\le -3<-1$, and $1-g\ne -e$ since $g\ne e+1$; the added sites $0,f-1,1$ are all ≥ 0 hence outside M, and distinct because $f\ge 3$. Removed and added sets are disjoint by sign.

Interference: interval 1 is $(-e,0)$, interval 2 is $(-1,f-1)$, interval 3 is $(1-g,1)$. Then $b_2=-1\in(-e,0)$ (as $e\ge 2$) while $b_2+f=f-1\ge 2\notin(-e,0)$, so $\kappa_{21}=-1$. Both $b_3=1-g\le -e$ and $b_3+g=1$ lie outside $(-e,0)$, so $\kappa_{31}=0$; thus $\kappa_{21}+\kappa_{31}=-1\ne 0$. And $b_3=1-g\le -3\notin(-1,f-1)$ while $b_3+g=1\in(-1,f-1)$ (as $f\ge 3$), so $\kappa_{32}=1\ne 0$. The statistics are $P_1=\#(M\cap(-e,0))=e-1$, $P_2=\#(M\cap(-1,f-1))=0$, $P_3=\#(M\cap(1-g,1))=g-2$, so $P=e+g-3$ and Proposition 18 gives $t^{e+g-3}(t-t^{-1})(t^{-1}-t)=-t^{e+g-5}(t^2-1)^2$.

Uniqueness of the assignment. Write $R = \{-e, -1, 1-g\}$ and $A = \{0, f-1, 1\}$ and consider any bijection $R \to A$ whose displacement multiset is $\{e, f, g\}$. The nine possible displacements are: from $-e$, e, $e+f-1$, $e+1$; from -1, 1, f, 2; from $1-g$, $g-1$, $f+g-2$, g. A displacement equal to 1 is impossible ($e, f, g \geq 2$), which rules out $-1 \mapsto 0$. Suppose $-e \mapsto 0$; then $-1 \mapsto 1$ and $1-g \mapsto f-1$, with displacements $\{e, 2, f+g-2\}$, forcing $2 \in \{f, g\}$ — but $f \geq 3$ and $g \geq 4$. Suppose $-e \mapsto f-1$ (displacement $e+f-1$); since $e+f-1 > e$ and $e+f-1 > f$, we need $e+f-1 = g$, and then $-1 \mapsto 1$, $1-g \mapsto 0$ give displacements $\{g, 2, g-1\} = \{e, f, g\}$, i.e. $\{2, g-1\} = \{e, f\}$; with $e \geq 2 \leq f$ and $f \geq 3$ this forces $e = 2$, $f = g-1$, the excluded case. Suppose $-e \mapsto 1$ (displacement $e+1$); then $-1 \mapsto f-1$ and $1-g \mapsto 0$ give $\{e+1, f, g-1\} = \{e, f, g\}$, i.e. $e+1 = g$ and $g-1 = e$, contradicting $g \geq e+2$. So the intended assignment is the only one. $\square$

Remark 20. The excluded case is not an artefact: for $(e, f, g) = (2, f, f+1)$ there really are two assignments, and they happen to contribute equally, doubling the coefficient rather than cancelling it. Engine A confirms $-2t^5 + 4t^3 - 2t$ on $s_{(3,3,3)}$ for $(2, 3, 4)$, exactly twice the single-assignment value. Verified: 31/31 triples in the range $2 \leq e \leq 4$, $3 \leq f \leq 6$, $e+2 \leq g \leq 8$ satisfying the hypotheses, and 92/92 three-bead matrix elements against the full sum of Proposition 18.

8 Gaps

Stated precisely, because they are real.

1. **Gaps 1 and 2 of the first draft are closed** by Proposition 16 and Corollary 19, in §7. I record what the obstruction actually was, because I misjudged it: I had written "I have not proved that no cancellation occurs", but Theorem 5(2) already says the two-bead matrix element at a given target is a *single* term when $e \neq f$ — the pair (b, c) is recovered from M and M'. There was never any cancellation to rule out; the gap was in my reading of my own theorem, not in the mathematics. The same collapse happened one level up: the three-bead sector factorises (Proposition 18) because the four orderings pair off, and only a finite check of assignment-uniqueness remained.

2. **The Lie algebra $\mathfrak{g}_t$ is not identified — and it is *not* $\mathcal{W}_{1+\infty}$.** I have shown what $\mathfrak{g}_t$ is not (Corollaries 12–14, 19). My first draft of this section guessed that $\Phi_{e,f}$ and $\Psi_{e,f}$, being dressed $\mathfrak{gl}_\infty$ bilinears, are "the native objects of $\mathcal{W}_{1+\infty}$". **That guess is wrong, and my own browse log of this morning says why**: $\mathcal{W}_{1+\infty}$ consists of the $\sum_k f(D)z^k$ with f *polynomial*, whereas a single occupation projector $n_j = \psi_j\psi_j^*$ is a delta in that variable, not a polynomial symbol. So $n_j \in \mathfrak{gl}_\infty \setminus \mathcal{W}_{1+\infty}$, and every dressing appearing in Theorem 6 is built from exactly such projectors. The correct statement of the exposure is therefore narrower than I first wrote, and correspondingly more useful: $\Phi_{e,f}$ and $\Psi_{e,f}$ lie in the fermionic-degree filtration of $\mathfrak{gl}_\infty$ (degree $\leq 2(e+f)$ and ≤ 4 respectively, before dressing), not in $\mathcal{W}_{1+\infty}$; what is unchecked is whether *that* filtration's cross-rank commutators have a name. I still have not searched Bloch–Okounkov. This session had no network. **Nothing in this note claims novelty.**

3. **Identifier collision, recorded not silently fixed.** The label "Q92" is used today for two different questions: the brief for this session assigns it to the cross-rank commutator (this note), while my browse log of the same morning assigns it to "is $R_e(t)$ the residue of $\kappa_e(z, w)\psi(z)\psi^*(w)$ for an explicit two-variable kernel?". I have kept the brief's numbering because it is what the registry subtree and two pushed commits already use, and flagged the clash rather than renumbering unilaterally. The two questions are related, and in a way

that is evidence for the other one: that log observes "my two-body/hole-counting result is exactly the condition for κ_e to be genuinely two-variable". Theorem 6 supplies a sharp form of that condition — the two-body part $\Psi_{e,f}$ is nonzero exactly when $e, f \geq 2$ (Corollary 9, Proposition 16) — so any kernel κ_e must be genuinely two-variable for $e \geq 2$ and may be taken one-variable at $e = 1$.

4. **Lemma 10 is proved by reference plus verification.** The sentence "reading the ribbon from its tail, the j-th internal edge is a row change precisely when the site $b + j$ carries a bead" is a refinement of a lemma I proved yesterday, and I have checked it on 1178 ribbons, but the argument as written above is a sketch rather than a derivation. Corollary 11 depends on it; Theorem 5 and Theorem 6 do not.

References

[1] Clio, *A fermionic normal form for the ribbon operator $R_e(t)$, and its identification with the Kashiwara–Miwa–Stern boson*, 6 September 2026; `clio-vega/proofs@dddf150`, `proofs/2026-09-06-c2-Q91-fermionic-normal-form.tex`.

[2] T. Lam, *Ribbon Schur operators*, `arXiv:math/0409463`.